DNA extraction v9.0 (modified BOMB)

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Abstract

DNA extraction using yttria-stabilized zirconia beads lysing and automated magnetic bead-based extraction.

Materials

1. Lysis master mix 870 ml (870 µl/sample)

TE buffer	225 µl
Lysis buffer	375 µl
10M Ammonium acetate	270 µl

2. TE buffer stocks 225 ml

1M Tris-HCl pH8.0	2.25 ml	10mM
1M EDTA	0.225 ml	1mM
ddH2O	adjust the volume with water to 225 ml	

Note

Prepare 1M Tris-HCl and 1M EDTA first.

Tris-HCl	7.88 g	50 ml
EDTA	14.61	50 ml

Note

adjust the pH of EDTA with solid NaOH or 1M NaOH to 8.0. The undissolved EDTA will be dissolved entirely when the pH reaches 8.0.

3. Lysis buffer stocks 375 ml

Guanidine thiocyanate (GITC)	177.3 g	4M
1M Tris HCl pH8.0	18.75 ml	50mM
Sodium Lauryl Sulfate	3.75 g	0.5 g
1M EDTA	7.5 ml	20mM
ddH2O	adjust the pH with HCl to 7.6–8.0 and the volume with water to 375 ml	

4. 10M Ammonium acetate 270 ml

Ammonium acetate	208.116 g
ddH2O	adjust the volume with water to 270 ml

Note

Ammonium acetate is hygroscopic. Do not add more than half of the water needed to dissolve it.

Procedure

Sample Collection

Step 1 *(30 s)*

Add **200 µL** of **0.5 mm beads** to a 2mL screw tube.



Step 2 *(30 s)*

Add **200 µL** of **1 mm beads** to a 2mL screw tube.



Step 3 *(30 s)*

Add**870 µL** Lysis master mix to 2mL screw tube. The final look:



Note

In 11F, 4°C fridge

Lysis master mix: **225 µL** of TE buffer + **375 µL** of lysis buffer + **270 µL** of 10M ammonium acetate

Step 4 *(1 min)*

Collect **20-50 mg** of **sample** to 2mL screw tube

Note

You can collect up to 100 mg of sample if you can until you bump into the low DNA quality or PCR success rate; by then it means too many inhibitors in the sample and you have to lower the input.

Sample lysis

Step 5 *(4 min)*

Put the 2mL screw tube on vortex for sample lysis, at 3200 rpm **4 min**

Note

Remember to balance if you have odd number of samples

Centrifugation

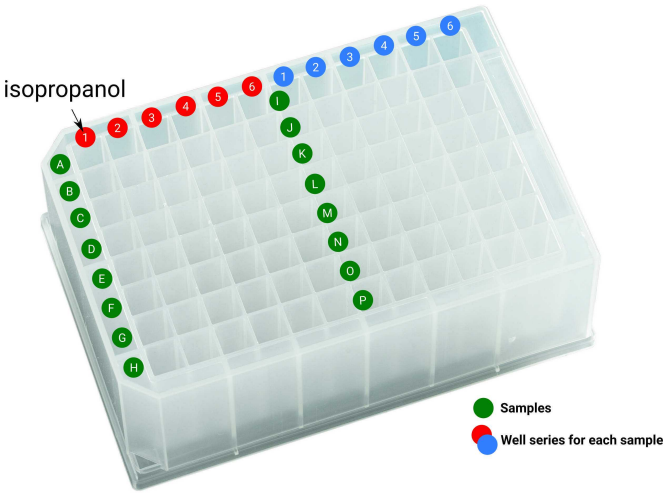
Step 6 *(3 min)*

Put 2mL screw tube in centrifuge for centrifugation, at this condition:**10 × g, 25 °C, 3 min**

DNA extraction

Step 7 *(30 s)*

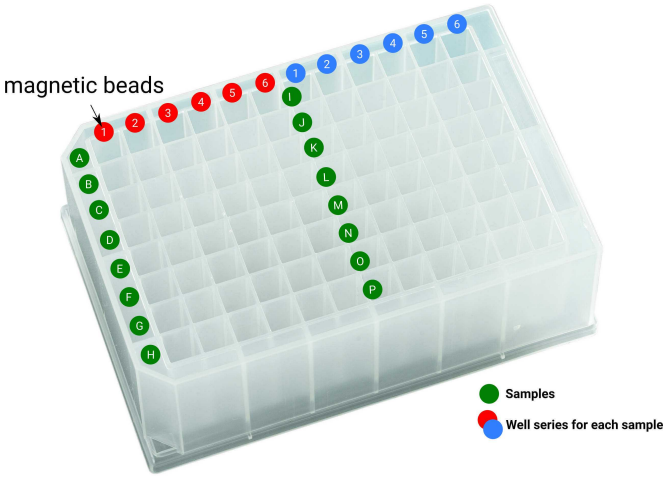
Add **350 µL** of **isopropanol** to the 1st well of 96 well plate



DNA purification

Step 0.21 *(30 s)*

Add **100 µL** of **magnetic beads** (10 mg/ml) to the 1st well of 96 deep well plate



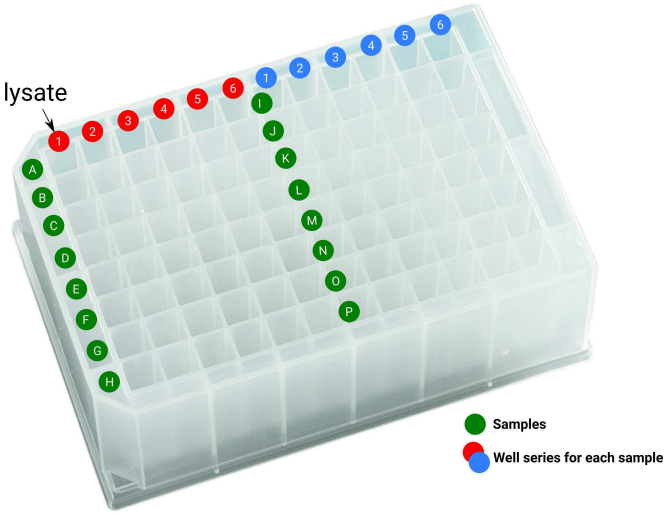
Note

Vortex the bottle and pipetting before using magnetic beads; re-do vortex after adding to 3 samples to prevent set down of magnetic beads

DNA extraction

Step 14 *(30 s)*

Add **300-500 µL** of the **sample (lysate)** from the 1.5mL centrifuged tube to the 1st well of 96 deep well plate

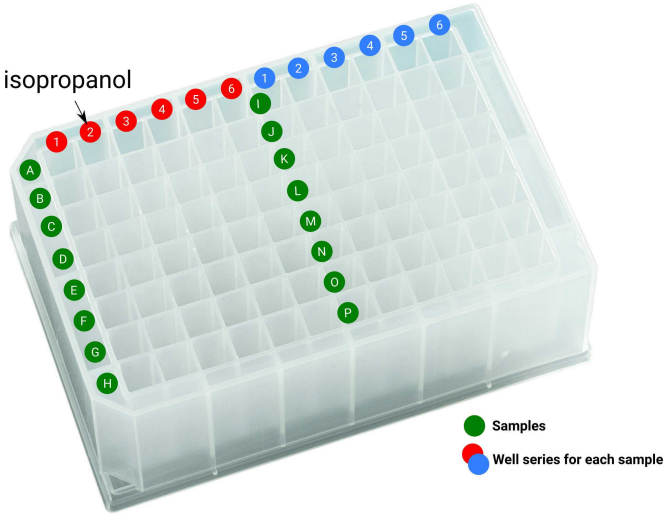


Note

Pipetting **as many lysate as you can**, as long as it's free of any cell debris (no solids in your tip)

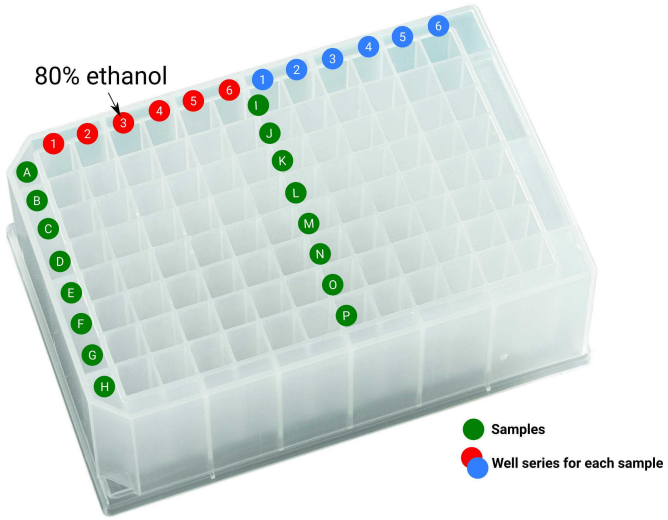
Step 9 *(30 s)*

Add **400 µL** of **isopropanol** to the 2nd well of 96 deep well plate



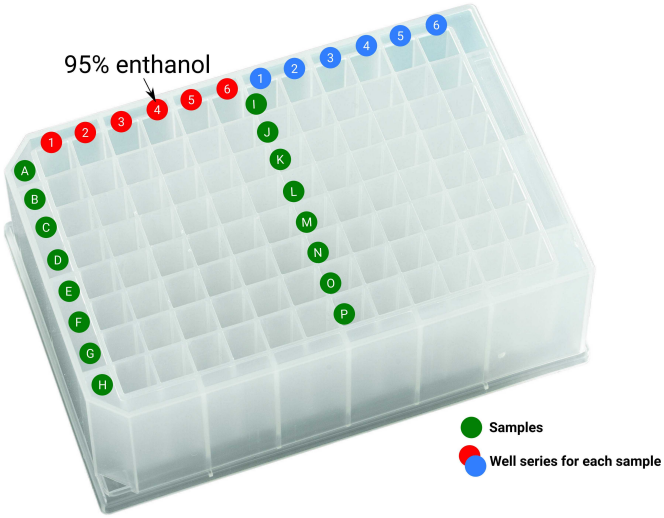
Step 10 *(30 s)*

Add **300 µL** of **80% ethanol** to the 3rd well of 96 deep well plate



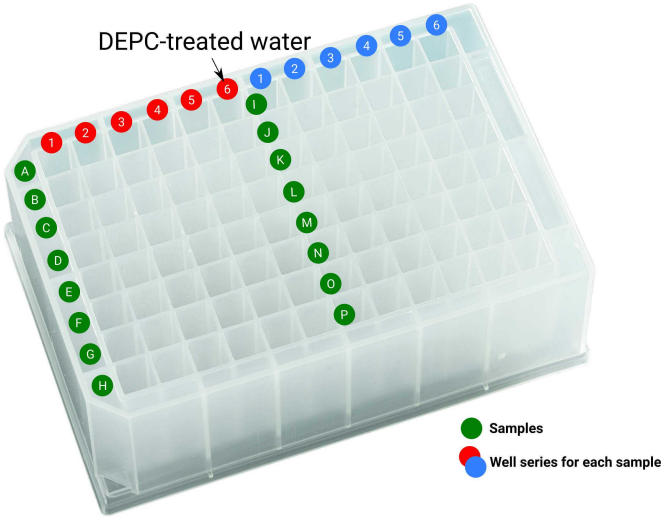
Step 11 *(30 s)*

Add **300 µL** of **95% ethanol** to the 4th well of 96 deep well plate



Step 13 *(30 s)*

Add **100 µL** of **DEPC-treated water** to the 6th well of 96 deep well plate



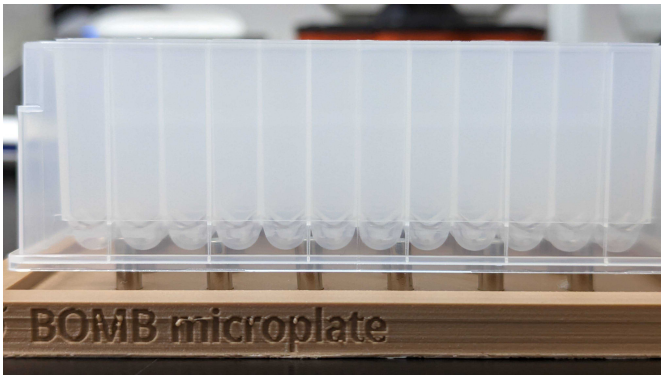
Step 15 *(34 min)*

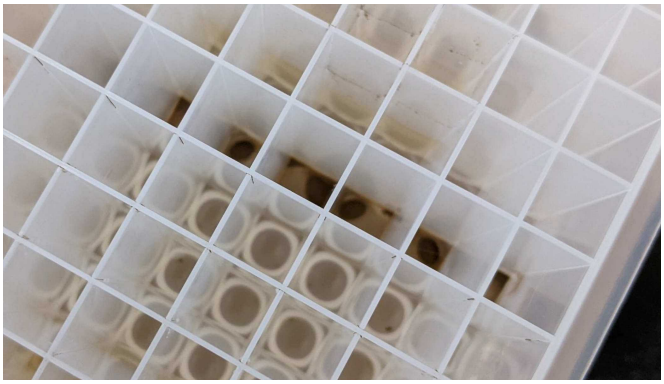
Put the prepared 96-deep well plate in the automated DNA extraction machine (ZiXpress 32) and set up the settings as below.

gDNA collection

Step 17

After the extraction is done, put on the 96 magnetic plate to pellet the magnetic bead residues.

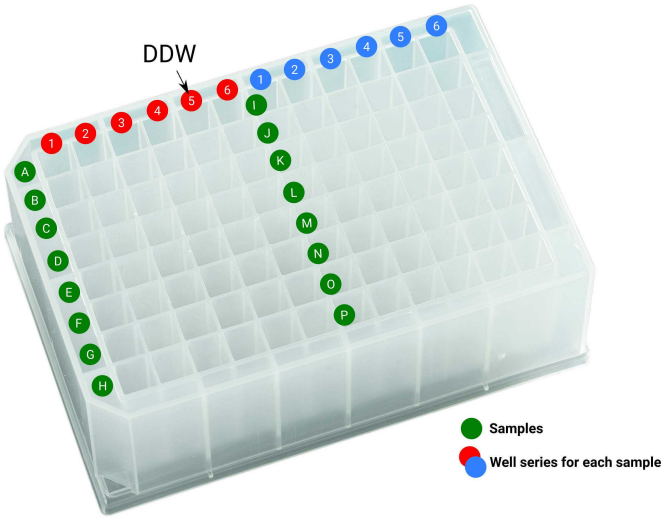




DNA extraction

Step 12

Add **300 µL** of **DDW** to the 5th well of 96 deep well plate



gDNA collection

Step 18

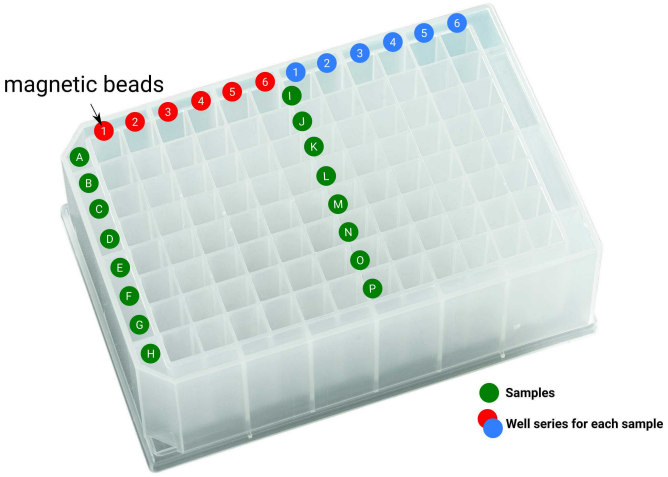
Collect **100 µL** of the **eluted sample** (avoid getting magnetic bead) as the DNA template for downstream experiments



DNA extraction

Step 8

Add **50 µL** of **magnetic beads (10mg/ml)** to the 1st well of 96 deep well plate



Program settings

Step 16

The automated extraction program settings for ZiXpress 32.

1	0	10	1000	3	120	0
2	0	2	400	3	60	0
3	0	2	300	3	60	0
4	15	0	0	0	0	0
4	0	2	300	3	60	0
5	0	1	300	3	0	0
6	0	5	100	3	120	55